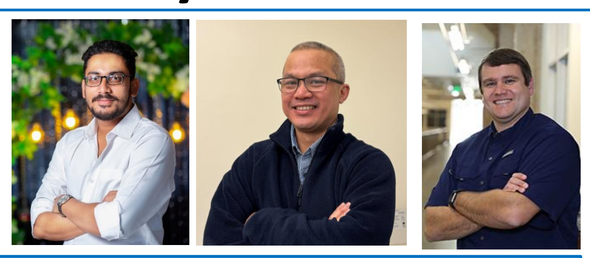


Failure Analysis of Geometrically Curved L-Shaped Carbon Fiber Composite Laminates



Kirtunia Rahul (PhD Student)
Dr. Arief Yudhanto
Dr. David Jack (Advisor)
Baylor University, Mechanical Engineering

Abstract: The objective of this research is to analyze the influence of relative ply orientation angle difference in stacking sequence on the failure of "L-shaped" composite parts under bending force. Samples of various angle differences from 15° to 90° have been fabricated and mechanically tested by four-point bending fixture to correlate the delamination with the increment of angle difference between adjacent plies. A finite element based predictive model is developed to predict the delamination onset for the "L-shaped" composite parts under bending moment. Finally, simulation results are compared to the experimental results in terms of load displacement behavior and fractographic images.

Introduction

In automotive and aerospace applications, Carbon Fiber Reinforced Plastic (CFRP) composites are utilized not only in flat panels but also in complex structures with curves, bends, spirals, and cylindrical shapes to meet specific design requirements. "L-shaped" brackets, featuring at least a 90° bend, are common examples of such intricate geometries. In aerospace engineering, these brackets are crucial for supporting and reinforcing various aircraft systems, functioning as stiffeners, crack stoppers, and fittings, all of which are subjected to significant stresses. However, during operation, these structures face substantial loading conditions, resulting in flexural moments and elevated stress levels that challenge their structural integrity. Ensuring the reliability of carbon fiber composite components in industrial settings necessitates addressing issues such as matrix fracture and delamination. ASTM D6415, a standard for determining curved beam strength (CBS) and delamination strength, is critical for assessing interlaminar tensile strength (ILTS) and delamination resistance. This research investigates the failure modes of "L-shaped" composite laminates under static four-point bending forces. A finite element analysis (FEA) model is developed to evaluate different ply stacking sequences with relative angle differences ranging from 0° to 90°, including unidirectional, helicoidal, quasi-isotropic, and cross-ply configurations. The study aims to enhance understanding of how these sequences affect the structural performance of CFRP components.

Manufacturing Process

- Manufacturing process is done by using two types of molds- male mold and female mold.
- Once all laminas are stacked, the female mold is set on the top of the male mold and the panel was cured using static press .

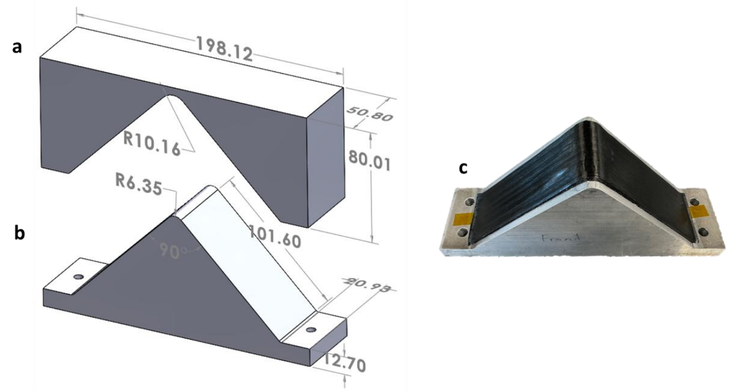


Figure 1: Mold to create "L-shaped" part (a) Female Mold, (b) Male Mold, (c) Prepregs laid on the male mold

Four different laminates of different relative angle difference between plies is manufactured for mechanical testing. The ply orientation of the laminates are listed as:

Laminate Name		Ply Orientation
Unidirectional	UD	$[0°]_{26}$
Helicoidal	HC	$[0°/15°/30°/45°/60°/75°/90°/-75°/-60°/-45°/-30°/15°/0]_S$
Quasi-isotropic	QI	$[0°/45°/90°/-45°/0°/45°/90°/-45°/0°/45°/90°/-45°/0]_S$
Cross-ply	CP	$[0°/90°/0°/90°/0°/90°/0°/90°/0°/90°/0°/90°/0°]_s$

Experimental Methods

- Four-point bending tests on L-shaped carbon/epoxy laminates were performed using an Instron 8801 (Norwood, MA, USA) with a 100 KN load cell .
- The top two rollers are placed at 60 mm apart and placed on the top of the sample where the bottom two rollers were set 100 mm apart.
- A constant crosshead speed of 2 mm/min was applied to the sample up to the failure.

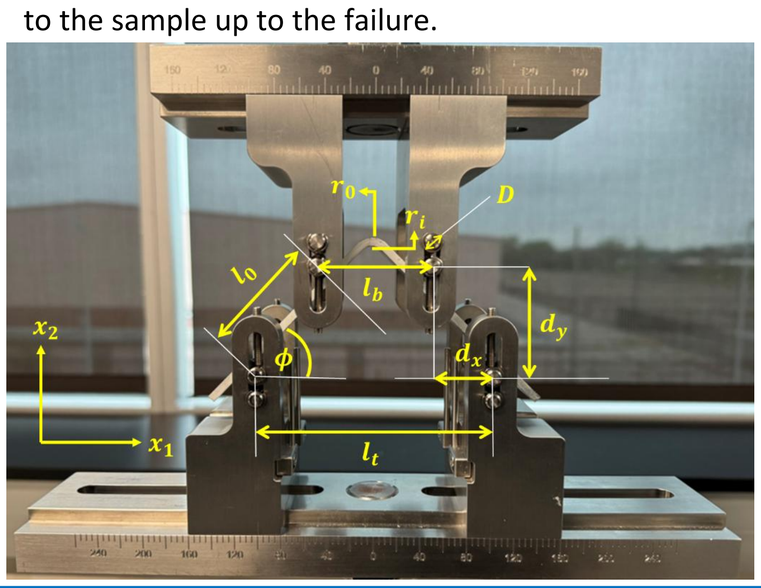


Figure 2: Four-point bending setup of "L-shaped" Composite

Quality Assessment and Improvement

- Thickness of a curved composite impacts the failure and stiffness of composite (figure 3(a)).
- Two spacers, each with a thickness of desired laminate thickness (in this instance, 3.9 mm), are inserted between the molds for constant thickness (figure 3(b)).

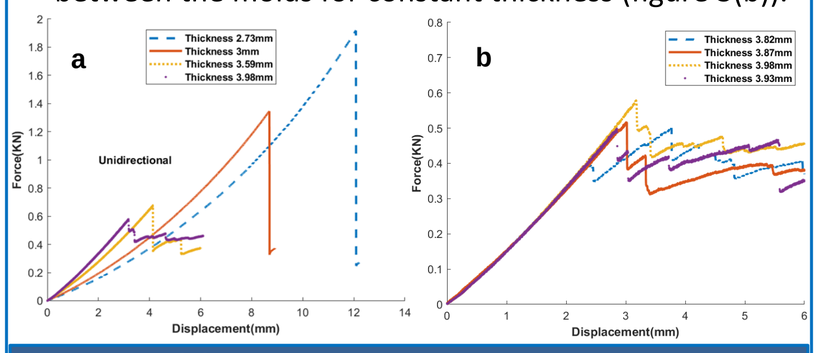


Figure 3: (a) Effect of thickness variation on different stacking orientation, (b) Four-point bending results for unidirectional samples after thickness correction

Effect of relative angle difference between plies

- With the increase of angle difference between plies, Ultimate load and failure displacement is increased.
- Because of more 0° plies, cross ply show less ultimate load and failure displacement than quasi-isotropic.

Name	Ultimate Load	Failure Displacement
Unidirectional	.54 KN	3.12 mm
Helicoidal	.64 KN	6.5 mm
Quasi-isotropic	.76 KN	7.65mm
Cross-ply	.71 KN	6.33 mm

Finite Element Method

- Created a custom Meso-scale 3D Model
- $\frac{1}{4}$ of the part was modeled for improved runtime.

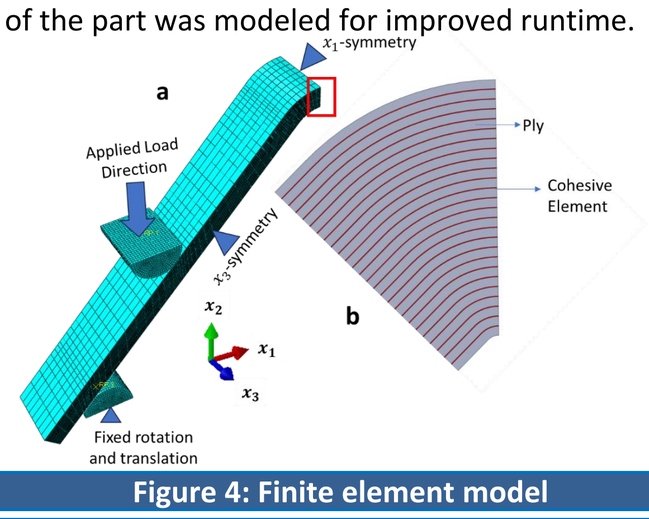


Figure 4: Finite element model

Results

- Interface failure location and depth is the same between the model and experimental observations
- For Unidirectional panels, first delamination was observed at the 11th and 12th interfaces. Subsequent delamination was noted between the 17th and 18th interfaces and between the 5th and 6th interfaces.

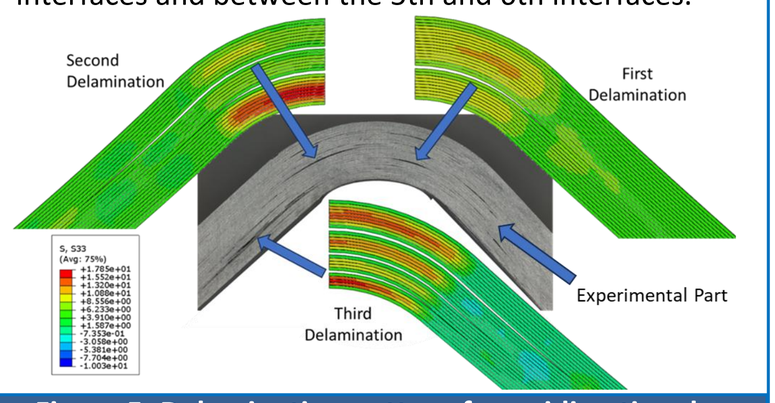
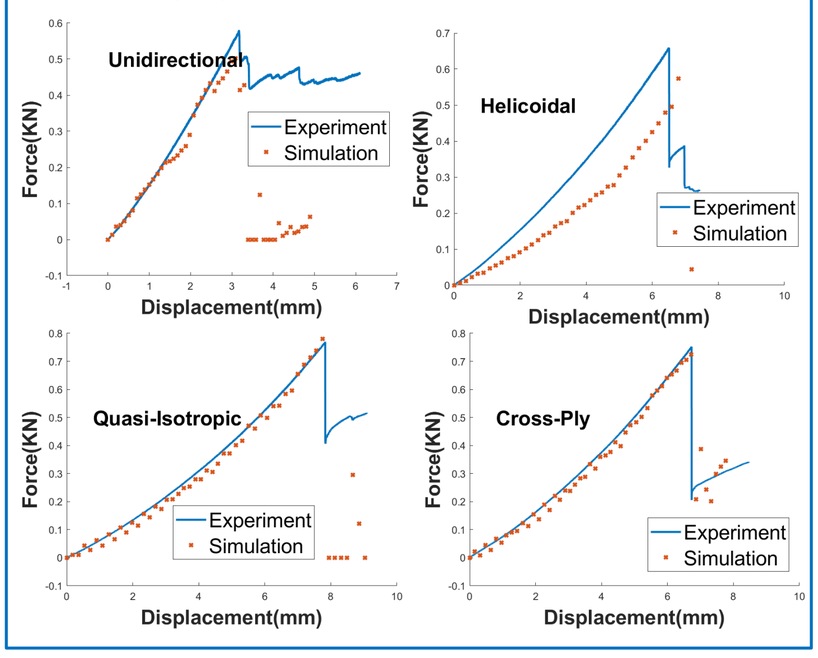


Figure 5: Delamination pattern for unidirectional composite

- Simulation show a close match with the experimental studies for all layup except helicoidal layup.
- Further study needs to be conducted for based on material properties for the simulation validation.



Summary and Conclusions: This research explored the impact of relative ply angle differences of ply orientation on the structural integrity of "L-shaped" Carbon Fiber Reinforced Plastic (CFRP) composites under bending forces. Mechanical testing and finite element analysis (FEA) revealed that 0° unidirectional laminates are prone to early delamination. The FEA model closely matched experimental results, providing valuable insights for improving the design and durability of CFRP components in aerospace and automotive applications. Dynamic loading conditions needs to be considered for further investigation of delamination pattern and life analysis.

Student Poster Competition Sponsored by DASSAULT SYSTEMES